Belief-based Inferences in Logical Reasoning

Zuleika Gasimova, Dr. Valerie Thompson

University of Saskatchewan

Introduction

Metacognitive monitoring[1] – self-assessment of thinking and progress on task

Metacognitive control[1] – strategic adjustment effort and time based on monitoring

Monitoring judgements are **based on cues**[2,3] available in the task environment (e.g. familiarity)

Experience-based inferences[2,4] – cues can give rise to subjective experiences of ease or difficulty (i.e. fluency)

Belief-based inferences[2] – people search for cues that might affect difficulty

Metacognitive illusions[5] – dissociations between metacognitive judgements and performance

Research Questions - Do people have beliefs how cues affect the difficulty of reasoning problems (Study 1), and do such cues influence monitoring and control (Study 2)?

Study 1

Which one of the problems below are more likely to be solved correctly?

Problem A	Problem B
All cooks are painters	All alizers are uplings
All painters are reporters	All uplings are spirots
All reporters are cooks	All spirots are alizers

A I Both I B

- Logic problems with familiar content create a subjective feeling of ease and lead to a false sense of certainty as opposed to unfamiliar content [6, 7, 8]

Do people have beliefs about how *familiarity* affects difficulty of logic problems?

- Participants saw two otherwise equivalent logic problems with either *familiar* or *nonword* terms for 10 ms
- After indicating which problem would be easier to solve, participants explained the basis of their judgement

Study 2

Are monitoring judgements sensitive to cue manipulations?

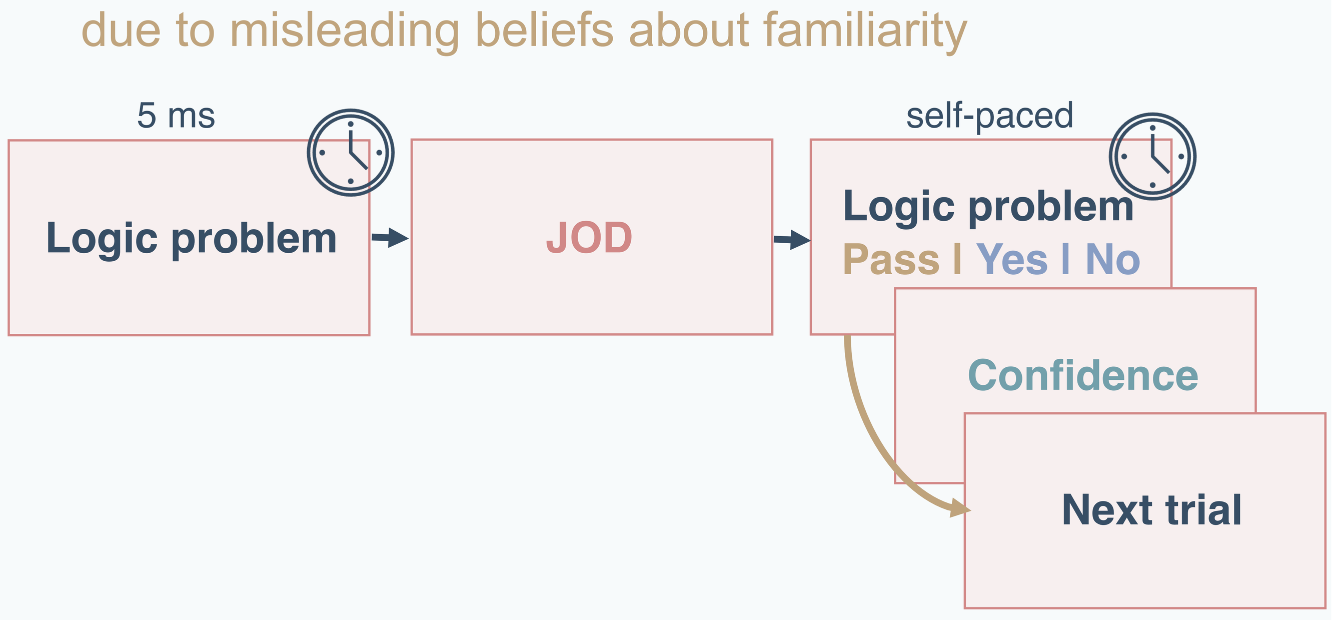
- Familiar terms will be (misleadingly) **judged as easier** than abstract terms.
- Familiar terms will elicit **higher confidence** than abstract terms.

Do people's monitoring judgements align with actual performance?

- There will be a **dissociation** between JoD and actual performance.

Are control behaviours sensitive to cue manipulations?

- People will **be more willing to give up** on abstract items due to misleading beliefs about familiarity



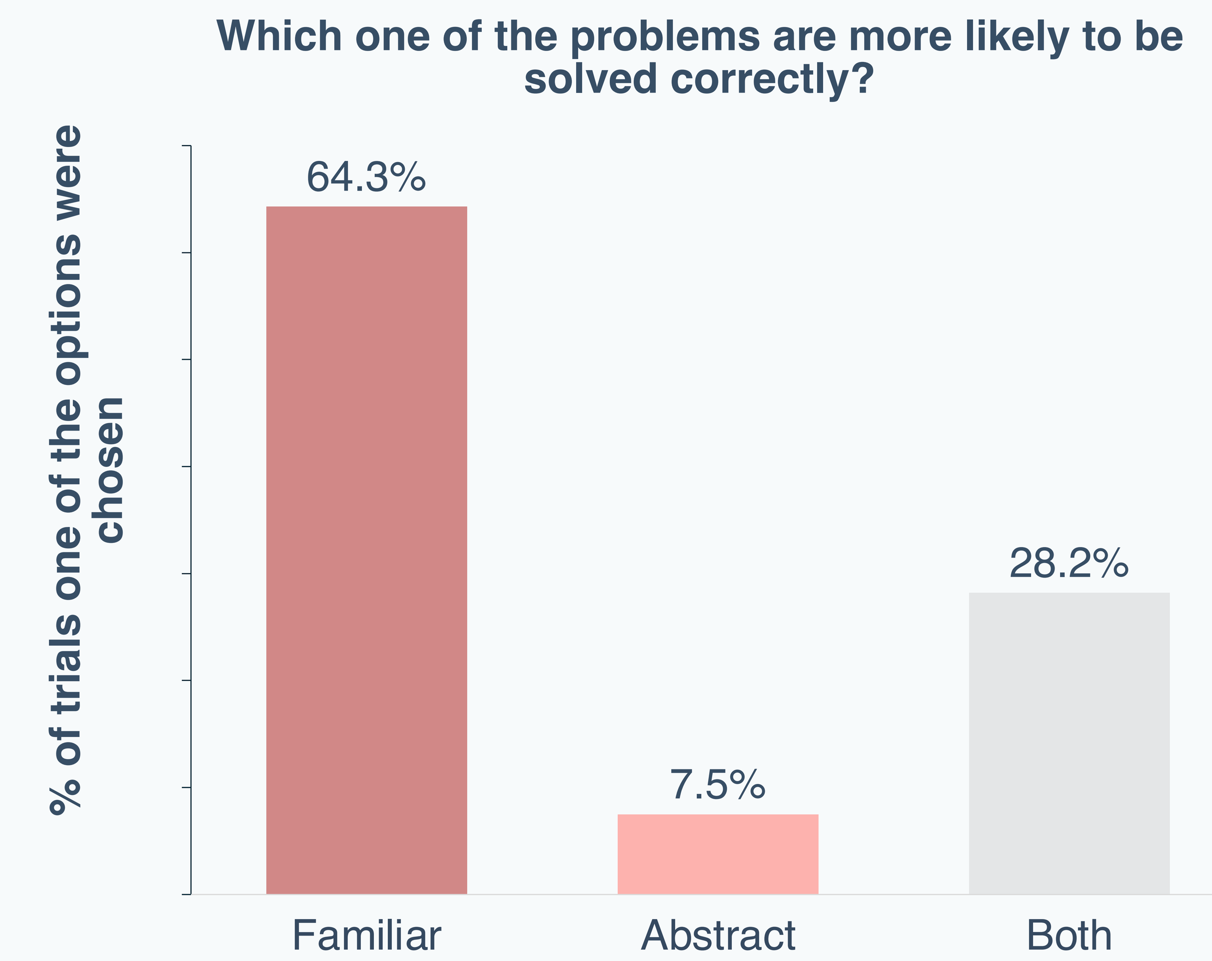
Results

Study 1 – Familiar problems are believed to be easier than abstract problems

N = 80 (M_{age} = 36.6, 52% female)

Participants articulated *(misleading)* beliefs about how familiarity affects the reasoning process:

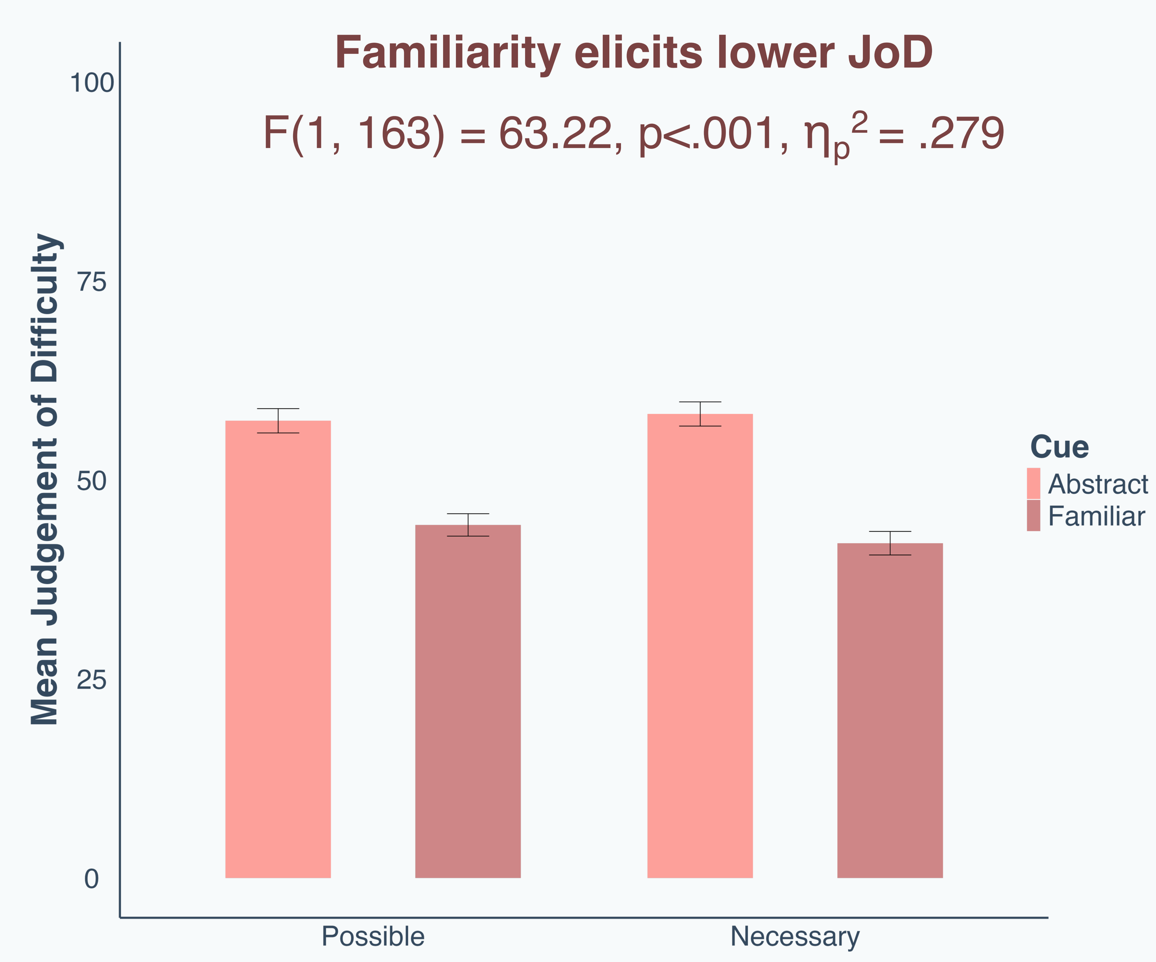
> *"Unfamiliar words create an obstacle to the thought process"*
>
> *"Familiar words make it so much easier to visualize the probable correct answer."*
>
> *"The unfamiliar words require far too long to make sense of the individual sentences given the short time allowed."*

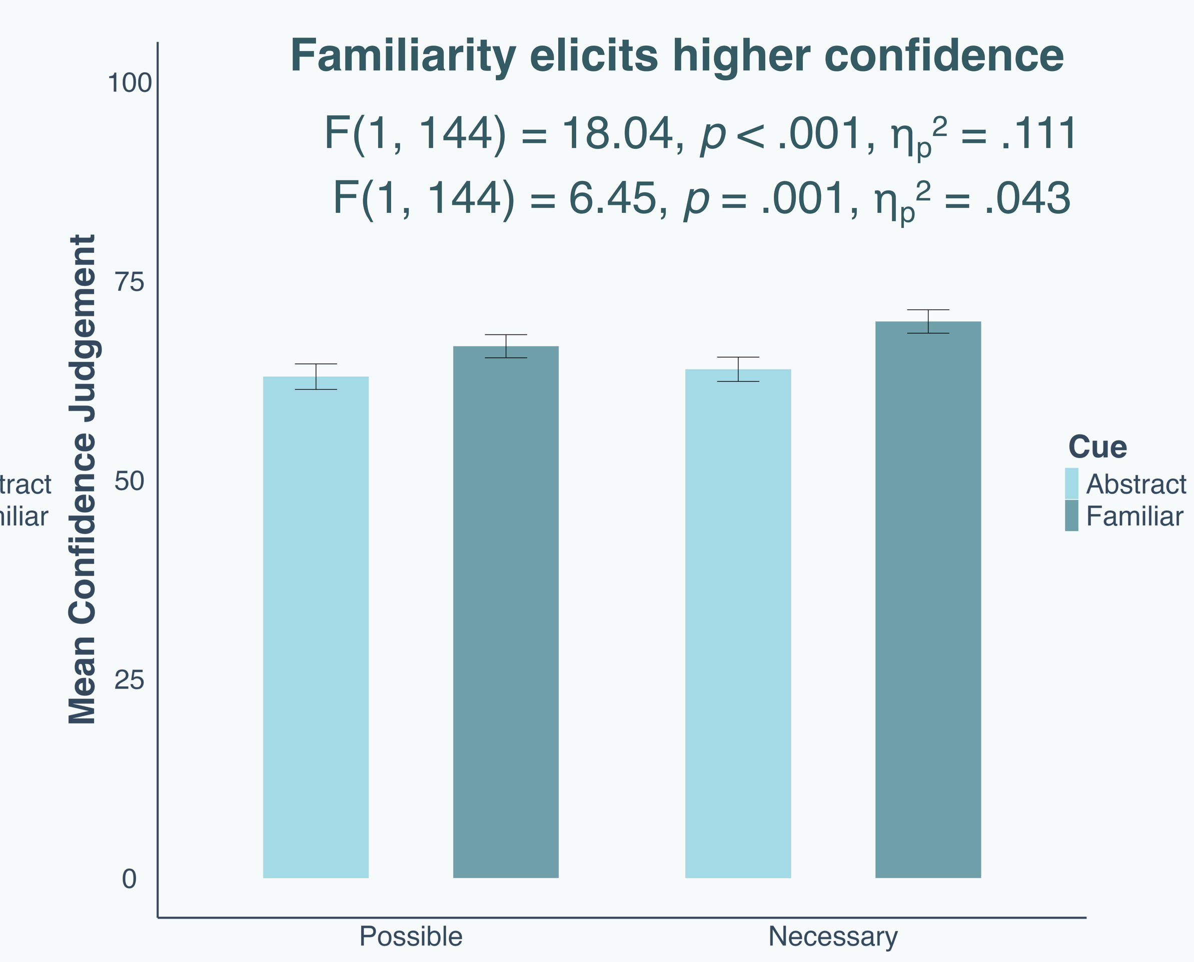


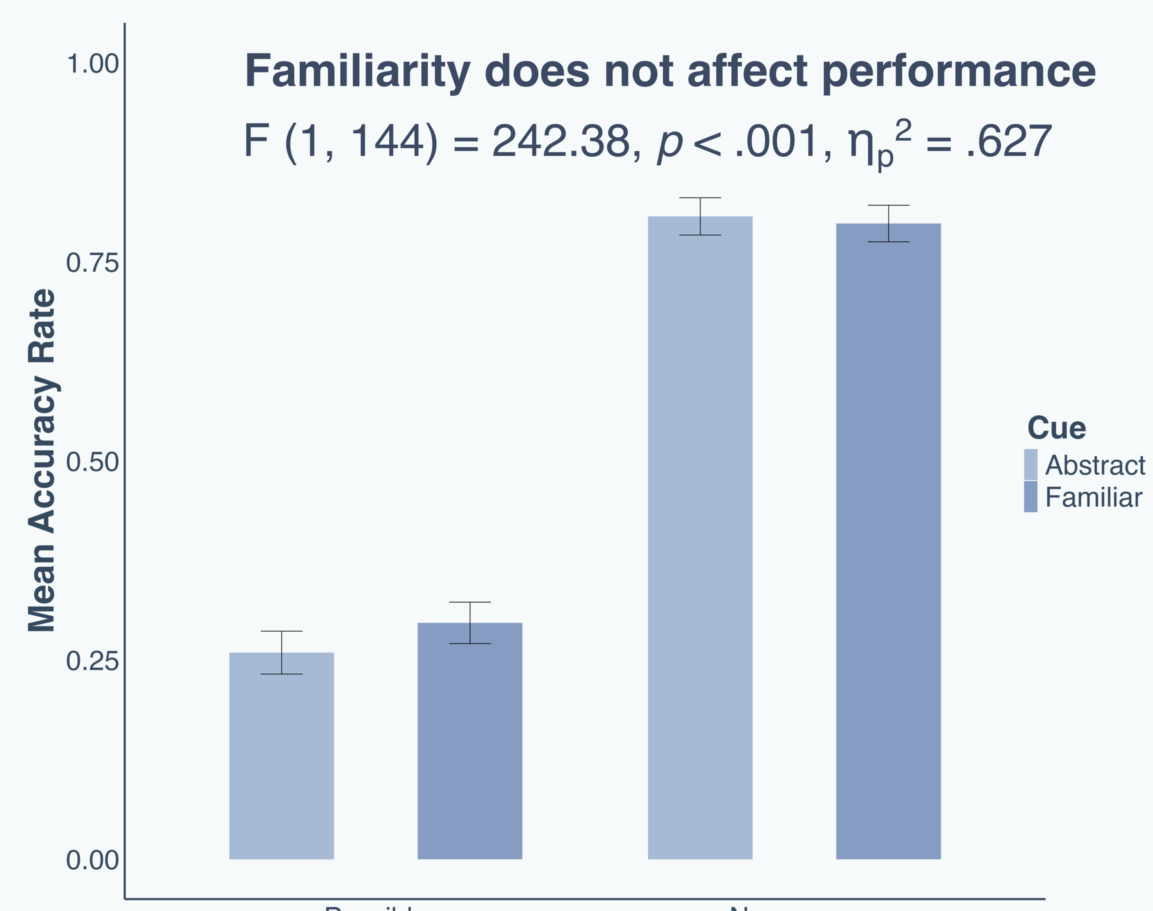
Study 2 – Familiarity affects monitoring & control

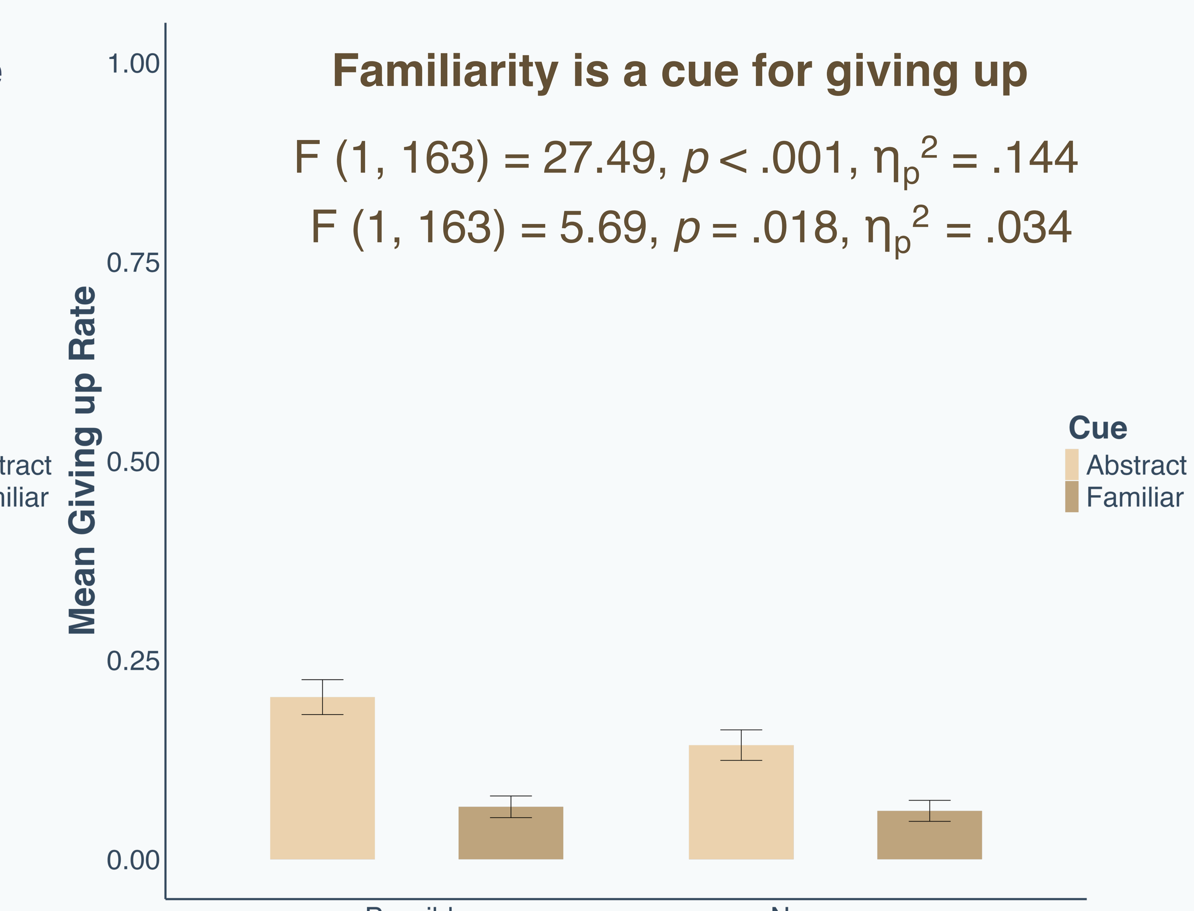
2 (Terms: Familiar, Abstract) x 2 (Validity: Necessary, Possible) within-subjects ANOVA

N = 178 (M_{age} = 34.2, 55% female)









Conclusion

Study 1 shows that people **may have or develop beliefs about how cues (familiarity) affect difficulty**

Study 2 showed that people use cues when making monitoring judgements (**JOD**, **Confidence**) and control decisions (**Giving up**).

Study 2 showed a dissociation between **difficulty judgements** and **performance** which misguides control decisions

Both studies show that **familiarity is a misleading cue** for difficulty of logic problems

References

1. Ackerman, R., & Thompson, V. A. (2017). *Trends in cognitive sciences*, *21*(8), 607-617.
2. Dunlosky, J., & Tauber, S. K. (2014). *Handbook of applied memory*, 444-464.
3. Koriat, A. (1997). *Journal of experimental psychology: General*, *126*(4), 349.
4. Shynkaruk, J. M., & Thompson, V. A. (2006). *Memory & cognition*, *34*(3), 619-632.
5. Undorf, M., Navarro-Báez, S., & Zimdahl, M. F. (2022). In *Cognitive Illusions* (pp. 307-323). Routledge
6. St BT Evans, J., Handley, S. J., Harper, C. N., & Johnson-Laird, P. N. (1999). *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *25*(6), 1495
7. Reder, L. M., & Ritter, F. E. (1992). *Journal of Experimental Psychology: Learning, memory, and cognition*, *18*(3), 435.
8. Markovits, H., Thompson, V. A., & Brisson, J. (2015). *Memory & cognition*, *43*, 681-693.

NSERC CRSNG